

Required functions

```
>>> def loadBoard(filename):
# loads the board from the file. It is returned as a list of lists
    board = []
    file = open(filename, "r")
    for line in file:
        row = line.strip().split()
        if row:
            board.append(row)
    file.close()
    return board

>>> def printBoard(board):
# prints the board to the console
    for row in board:
        print(" ".join(row))

>>> def possibleMoves(position, board):
# calculates and returns the positions from a given position
    x, y = position
    moves = []
    n = len(board)

    for dx in [-1, 0, 1]:
        for dy in [-1, 0, 1]:
            if dx == 0 and dy == 0:
                continue
            newX, newY = x + dx, y + dy
            if 0 <= newX < n and 0 <= newY < n:
                moves.append((newX, newY))
    return moves
```

Code testing

```
>>> myBoard = loadBoard("board.txt")
```

```
>>> printBoard(myBoard)
```

```
J O P Y
```

```
C M P V
```

```
X F E G
```

```
P G V U
```

```
>>> possibleMoves((0,0), myBoard)
```

```
[(0,1), (1,0), (1,1)]
```

```
>>> possibleMoves((2,2), myBoard)
```

```
[(1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (3,3)]
```